Technology Stack

Date	30 Jume 2026
Team ID	N/A
Project Name	Personalized Networking Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	Streamlit (Python)	Provides a simple and interactive user interface where users can enter event details, generate networking conversation starters, perform fact-checking, and view conversation history with minimal development effort.

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	FastAPI (Python)	Handles API requests, connects AI models and external services, processes user inputs efficiently, and provides high-performance REST APIs with automatic Swagger documentation.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	JSON Files (history.json & feedback.json)	Stores conversation history and user feedback locally. Lightweight, easy to maintain, and suitable for local deployment without requiring a database server.
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	Local Deployment using Uvicorn & Streamlit	Runs FastAPI backend using Uvicorn and Streamlit frontend locally for development, testing, and demonstration without cloud infrastructure.

5	**Version Control & CI/CD**	Git & GitHub	Tracks source code changes, enables collaboration among team members, maintains version history, and supports project management.
	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	Hugging Face Transformers (DistilBERT & GPT-2), Wikipedia API, Pytest, httpx TestClient	DistilBERT performs topic classification, GPT-2 generates personalized conversation starters, Wikipedia API provides fact-checking, while Pytest and httpx TestClient are used for unit and integration testing.